

Expected Value and Variance

CSU, Chico Math 350

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Concept

Probability mass functions provide a global overview of a random variable's behavior. Many times we don't need to know everything about a variable, but we often want to summarize it in some way.

One feature of a distribution which we might be interested in is the central tendency of a variable. One measure of central tendency is the **expected value** or **mean** of the observation. The terms expected value and mean can be used interchangeably.

Definition

For a discrete random variable X with a PMF $p(x)$, the expected value of X is

$$E(X) = \sum_x x * p(x)$$

where the PMF $p(x) = P(X = x)$.

In other words, the expected value of X is the sum of each of its possible values x times the value's probability.

Example 1

Two books are assigned for a statistics class: a textbook costing \$137 and its corresponding study guide costing \$33. The university bookstore determined 20% of enrolled students do not buy either book, 55% buy the textbook only, and 25% buy both books, and these percentages are relatively constant from one term to another. Let X be a random variable that denotes how much a single

student will spend on their course materials. The PMF is:

* Textbook only: \$137

* Textbook + Study Guide: $\$137 + \$33 = \$170$

* Neither: \$0

x	0	137	170
$p(x)$	0.20	0.55	0.25

Example 1, Theoretical

Calculate $E(X)$ and interpret this value in context:

```
0 * .20 + 137 *.55 + 170*.25  
## [1] 117.85
```

On average, a single student will spend \$117.85 on their course materials.

Example 1, Simulated

Confirm your results using simulation.

```
dollars <- c(0, 137, 170)
prob <- c(.2, .55, .25)

spend <- sample(dollars, size=10000,
prob=prob, replace=TRUE)
mean(spend)
## [1] 117.5886
```


Exercise 1

A retirement portfolio's value increases by 18% during a financial boom and by 9% during normal times. It decreases by 12% during a recession. What is the expected return on this portfolio if each scenario is equally likely?

STEPS

1. Determine the PMF (what would the table look like?)
2. Calculate the theoretical expected value. Write your answer in a full sentence in context of the problem.
3. Confirm using simulation.

x			
$p(x)$			

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Concept

Although the mean is a useful descriptive statistic, it only gives us an idea of where the center of the distribution is located. For instance, the following table gives the monthly temperature of New York City and San Francisco:

<i>months</i>	J	F	M	A	M	J	J	A	S	O	N	D
<i>NYC</i>	32	34	42	53	63	72	77	76	68	57	48	37
<i>SF</i>	49	52	53	56	58	62	63	64	65	61	55	49

The mean temperature for San Francisco is about 57 degrees and the mean temperature for New York is around 55 degrees. So, their mean yearly temperature is about the same. Do you notice anything different about the two cities with regards to monthly temperatures?

Definition - Variance

Suppose random variable X has the mean $\mu = E(X)$. The variance of X , denoted by $Var(X)$ or σ^2 , is defined:

$$\begin{aligned} Var(X) &= \sigma^2 = E[(X - \mu)^2] \\ &= \sum_x (x - \mu)^2 * p(x) \end{aligned}$$

In other words, the variance of X is the sum of the squared differences of each possible value x and the mean μ , times the value's probability. It measures the spread or dispersion of the distribution around its mean.

This can also be written:

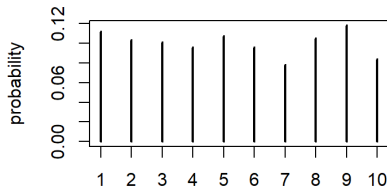
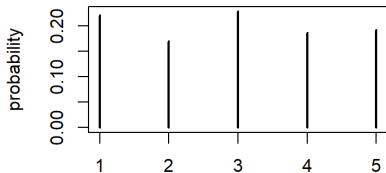
$$Var(X) = E[X^2] - E[X]^2$$

Definition - Standard Deviation

The standard deviation of a random variable X , $SD(X)$, is the square root of the variance. We denote the standard deviation by σ and the variance by σ^2 .

Which of the two distributions below has larger variance?

```
par(mfrow=c(1,2))
plot(proportions(table(sample(1:5, size=1000,
replace=TRUE)))), ylab="probability")
plot(proportions(table(sample(1:10, size=1000,
replace=TRUE)))), ylab="probability")
```



Example 2: Textbooks Revisited

Let's return to Example 1 and calculate $\text{Var}(X)$ and $\text{SD}(X)$.

Reminder: The textbook costs \$137, the study guide costing \$33. 20% of students don't buy either book, 55% buy the textbook only, and 25% buy both books.

```
x <- c(0, 137, 170)
p.x <- c(.2, .55, .25)
mu <- sum(x*p.x)
```

THEORETICAL

```
(x.minus.mu <- x - mu)
## [1] -117.85    19.15    52.15
(var.dollars <- sum(x.minus.mu^2 * p.x))
## [1] 3659.327
sqrt(var.dollars)
## [1] 60.49238
```

Example 2: Textbooks Revisited

SIMULATION

```
spend <- sample(x, size=10000, prob=p.x, replace=TRUE)
var(spend)
## [1] 3595.947
```

Bonus: You may have noticed that the formula for $\sum_x (x - \mu)^2 * p(x)$ has the same format as a dot product. You can perform vector multiplication like this in R using the `%*%` operator.

```
x.minus.mu^2 %*% p.x
##           [,1]
## [1,] 3659.327
```

Exercise 2

Return to the retirement portfolio question (*recap: the value increases by 18% during a financial boom and by 9% during normal times, and decreases by 12% during a recession. Each scenario is equally likely*). Calculate the variance and standard deviation.

STEPS

1. Reuse the PMF from before to define probabilities
2. Calculate the theoretical value and write your answer in a full sentence in context of the problem.
3. Confirm using simulation.

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.